Mode Choice For The Intermodal Transportation Considering Carbon Emissions

Jun Zhang, Hong Wei Ding, Xiao Qing Wang, Wen Jun Yin, Tian Zhi Zhao, Jin Dong
IBM Research - China
Beijing, China
zjuncrl@cn.ibm.com

***Abstract*—The intermodal transportation as an effective logistics solution can provide timely and high quality service for customers. Especially with the rapid development of logistics industry, the intermodal transportation is becoming a very common practice in the industry today. The operation cost can be reduced to a great extent if different transportation modes are well connected and synchronized. Most existing research about intermodal transportation optimization focused on minimizing the total logistics cost, and/or maximizing the customer satisfied degree, etc. The impact of carbon emissions by different transportation carriers is seldom considered. However, transportation industry plays an important role for the carbon emissions reduction. Some corresponding rules, such as carbon tax, have been proposed to help reducing the carbon emissions in developed countries. Logistics companies need to balance the profit and the carbon emissions. This paper presents an approach to optimize the choice of transportation modes for intermodal transportation with consideration of carbon emissions. An integer programming model is proposed, of which the objective is to minimize the total cost including both logistics costs and costs related to carbon emissions. The computational results are also presented to illustrate the impact of considering carbon emissions on intermodal transportation.**

Keywords- intermodal transportation;carbon emissions

I. INTRODUCTION

With the rapid development of modern economy, the speed of the material flow and the product flow became quicker than before. The logistics industry is facing greater challenge on both capacity and efficiency. Moreover, the economic globalization requires logistics enterprises to move material and products across wide regions in very limited time. Single transportation mode cannot satisfy the demand. The intermodal transportation, as an efficient transportation method, is adopted widely by many logistics enterprises. It can integrate different kinds of transport (air, water, rail and road) to complete transit assignment. The flexible combination of different transport can guarantee the time and take the advantage of different transport well to improve competitive power of the enterprises. Some factors need to be considered during the process of intermodal transportation. For example, the choice of transport in each trip, the transfer point, the transportation time, the route, the transportation quality and the cost. The reasonable decision of those factors can help logistics enterprises gain more profit which is the first concerned by the enterprises. Some researches have been done to help the intermodal transportation be more effective. The combined-mode choice is a central decision. The chance-constrained goal programming model proposed by Min [1] and integer programming model proposed by Reddy and Kasilingam [2] are used to decide it with the objective of minimizing the cost. Moreover, the terminal location is not a negligible factor. Van Duin and Van Ham [3] presented a three-stage approach to decide the location of the intermodal terminal. Arnold *et al*. [4] proposed a linear 0-1 programming model to optimize the terminal location. Chang [5] addressed a intermodal multicommodity flow model with multi-objective. Time windows, transportation mode and economies of scale are all considered in the model to get the best transportation routes. Besides, several reviews are presented. Bontekoning *et al*. [6] made a detailed review about intermodal rail–truck freight transportation. The different definitions of intermodal transportation from literatures reviewed are listed. The intermodal rail–truck freight transport problem is divided into eight categories to analyze the characteristics of each category. Macharis and Bontekoning [7] gave a review about existing models in the field of intermodal transportation.

As the logistics activities increasing, huge pressures are imposed on the environment. The abundant greenhouse gases generated by logistics activities have affected the climate. More natural calamities take place for climate-change. Hence, reducing carbon emissions is advocated and compelled by many governments in the world. Logistics activities are regarded as playing a significant role to reduce carbon emissions. Low carbon will be a challenge for all logistics enterprises in the near future. Carbon emissions occurred among each stage of the supply chain. The transportation and inventory belongs to the delivery stage of supply chain, which would consume the energy and emit carbon dioxide. The researches about carbon emissions associated with transportation, inventory or supply chain have emerged in recent years. Ahn and Rakha [8] investigated the carbon emissions by microscopic and macroscopic tool to find the relationship between the route choice and the carbon emissions. Ren *et al*. [9] developed a multi-objective model to optimize the operating strategy of distributed energy resource systems with assessing and minimizing carbon emissions. Cholette and Venkat [10] studied the energy and carbon emissions of logistics in wine industry to analyze the factors which affect the amount of energy and carbon emissions.

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Sundarakani *et al.* [11] presented a carbon footprint modeling to evaluate the effect of carbon emissions on supply chain.

The transportation and inventory are the main activities of intermodal transportation, therefore the factor of carbon emissions need to be considered in the process of intermodal transportation. However, few researches of intermodal transportation consider it. Liao *et al.* [12] proposed an activity-based emissions modeling to evaluate the carbon emissions by truck. The conclusion shows that reasonable choosing transportation mode can reduce the cost and carbon emissions. Kim *et al.* [13] developed a multi-objective model to balance the carbon emissions and the logistics costs in given intermodal and truck-only freight networks. The results show that the trade-off between the carbon emissions and the logistics costs will be broken by the demand and capacity changing.

The carbon emissions are becoming an obligatory factor which needs to be considered in the modeling. The amount of carbon emissions can affect the choice of transportation carriers. Hence, the model considering carbon emissions is proposed in this paper to choose a kind of low carbon combined-mode to complete transportation. The objective is to minimize the total operation costs of the intermodal transportation. The operation costs include transportation cost, transfer cost, penalty cost and the cost incurred by carbon emissions.

Following the introduction, we give a detailed description about the optimization problem in section 2. An integer programming model will be proposed in section 3. In section 4, the computational experiment will be shown to verify the validity of the modeling. The result analyses are also given in this part. In section 5, the conclusion is presented.

II. PROBLEM DESCRIPTION

The intermodal transportation problem usually considered how to choose the combined-mode and route to minimize the total costs. However, the carbon emissions are seldom considered in the model. As we know, greenhouse gas damage the environment and people's health. It cannot be neglected during the process of optimization. Therefore, the carbon emissions are described in the proposed model.

Considering one logistics enterprises will deliver a batch of products from city A to city B. Several cities will be transited during the transportation. There are several transportation modes existing from A to B. The problem can be described by one virtual network shown in Fig.1. A and B represent the origin and destination respectively. The red rectangles represent the transfer cities which must be passed from A to B. The white points are the different transportation modes. One transportation mode needs to be chosen on each trip. Suppose the products can be transferred on these transfer cities, but the products cannot be split. The transportation mode can be and only can be changed once on each transfer point. The transfer activity will incur transfer cost and consume time. Suppose the transportation demand can be satisfied by any transportation mode. The differences among the several transportation modes are transport time, cost, carbon emissions and convenience degree. The delivery time of the entire transportation has been limited. If the products are delivered to B earlier than the given

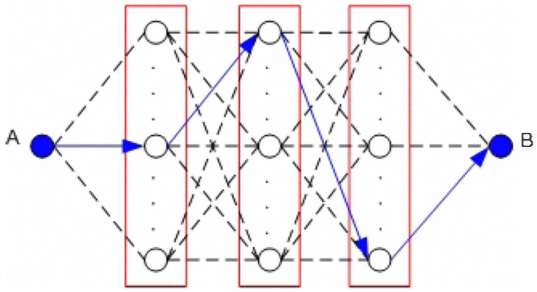


Figure 1. The virtual network of intermodal transportation

time, the inventory costs will be charged. Contrarily, the penalty cost will be charged when the products arrive late. As we know, the transportation needs to consume fuel and generate carbon emissions. The carbon emissions also exist in the process of store for consuming the electric energy. Hence, the process of transfer also emits carbon by transitory transportation and inventory. The entire carbon emissions include the emissions of transportation, transfer and inventory.

The enterprise will decide the transportation mode of each trip to minimize the total costs and the carbon emissions. The costs include the transportation cost, transfer cost, penalty cost and carbon emissions cost. The transfer cost is incurred by the transfer fee, temporal transportation cost and inventory cost. An integer programming model is proposed to describe the problem.

III. MATHEMATICAL MODEL

The proposed intermodal transportation model considered the carbon emissions and some constraints from Reddy and Kasilingam's model [2]. Following the notation given in this section, the mathematical model of the proposed problem is presented.

A. Notations

To model the problem, the following notations are defined.

1) Set

V represents the set of all cities except destination;

V_1 represents the set of all cities except origin and destination;

N represents the whole transportation mode;

2) Index

i,j=index of the cities, $i, j \in V$;

k,l= index of the transportation mode, $k,l \in N$;

3) Parameter

$d_{i,i+1}^{k}$ = the distance from city i to city $i+1$ by mode k ;

c_k = the cost coefficient by mode k per unit distance per unit weight;

$c_i^{k,l}$ =the cost coefficient of changing the mode k to l on city i per unit weight;

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C_t = the cost coefficient of carbon emissions;

$t_{i,i+1}^k$ =the delivery time of delivering products from i to $i+1$ by mode k ;

$s_i^{k,l}$ =the transfer time of changing the mode k to l on city i per unit weight;

α =the coefficient of inventory cost of earlier arriving per unit time per unit weight;

β =the coefficient of penalty cost of later arriving per unit time per unit weight;

g^k =the coefficient of carbon emissions per unit distance per unit weight by mode k ;

$h_i^{k,l}$ = the coefficient of carbon emissions per unit weight on city i when change mode from k to l ;

P =the coefficient of carbon emissions generated by inventory per unit weight per unit time;

M =the weight of the batch of products;

T_1 =the upper bound of the time required;

T_2 =the lower bound of the time required;

O =the limited amount of carbon emissions

4) Variables

t =the total delivery time;

$h(t)$ =the inventory cost;

$p(t)$ =the penalty cost;

5) The decision variables

$$x_{i,i+1}^k = \begin{cases} 1 & From\ i\ to\ i+1\ by\ \text{mode}\ k \\ 0 & otherwise \end{cases};$$

$$y_i^{k,l} = \begin{cases} 1 & the\ \text{mode}\ changed\ from\ k\ to\ l\ on\ city\ i, k \neq l \\ 0 & otherwise \end{cases};$$

B. Mathematical Model

The model is formulated as follows:

$$\sum_{i\in V}\sum_{k\in N} c_k d_{i,i+1}^k x_{i,i+1}^k M + \sum_{i\in V_1}\sum_{k\in N}\sum_{l\in N} c_i^{k,l} y_i^{k,l} M + p(t) + h(t)$$
$$+C_t\Big(\sum_{i\in V}\sum_{k\in N} g^k d_{i,i+1}^k x_{i,i+1}^k M + \sum_{i\in V_1}\sum_{k\in N}\sum_{l\in N} h_i^{k,l} y_i^{k,l} M + PM(T_1 - t)\Big), \quad (1)$$

S.t.

$$\sum_{k\in N} x_{i,i+1}^k = 1 \quad \forall i \in V, \quad (2)$$

$$\sum_{k\in N}\sum_{l\in N} y_i^{k,l} = 1 \quad \forall i \in V_1, \quad (3)$$

$$x_{i-1,i}^k + x_{i,i+1}^l \geq 2y_i^{k,l} \quad \forall i \in V_1, k \in N, l \in N, \quad (4)$$

$$\sum_{i\in V}\sum_{k\in N} x_{i,i+1}^k t_{i,i+1}^k + \sum_{i\in V_1}\sum_{k\in N}\sum_{l\in N} y_i^{k,l} s_i^{k,l} = t, \quad (5)$$

$$p(t) = \begin{cases} \beta \cdot M \cdot (t - T_2) & t > T_2 \\ 0 & t \leq T_2 \end{cases}, \quad (6)$$

$$h(t) = \begin{cases} \alpha \cdot M \cdot (T_1 - t) & t < T_1 \\ 0 & t \geq T_1 \end{cases}, \quad (7)$$

$$\sum_{i\in V}\sum_{k\in N} g^k d_{i,i+1}^k x_{i,i+1}^k M + \sum_{i\in V_1}\sum_{k\in N}\sum_{l\in N} h_i^{k,l} y_i^{k,l} M + PM(T_1 - t) < O, \quad (8)$$

$$x_{i,i+1}^k, y_i^{k,l} \in \{0,1\} \quad \forall i \in V, k \in N, l \in N. \quad (9)$$

The objective function is to minimize the total costs which include five terms. The first term is transportation cost and the second term is the transfer cost. The third and forth terms are the penalty cost. The last term is the costs of total carbon emissions. This term is composed by three parts. The first part is the carbon emissions generated by transportation. The second part is the carbon emissions by transfer. The last part is the carbon emissions generated by the inventory for earlier arrival. Constraint (2) is to guarantee there is only one kind of transportation carrier to complete the transportation between two cities. Constraint (3) is to make the transfer from one transportation mode to another mode on one city be only once. Constraint (4) guarantees the continuity of changing transportation mode. Equation (5) represents the delivery time. Constraints (6) and (7) state the inventory and penalty costs. Constraint (8) limits the total amount of carbon emissions.

The proposed model is a 0-1 integer programming model. IBM ILOG CPLEX[14] can be used to solve this problem effectively. The computational experiment will be given in next section.

IV. COMPUTATIONAL EXPERIMENT

To verify the proposed model, one computational experiment is given. An imaginary scenario is addressed based on the hypothetical data.

A. The Experiment Data of The Parameters

Considering one logistics enterprise will delivery a batch of products from city A to city B, there are eight cities between A and B represented by C to J. The cities C to cities J will be passed in sequence when the products are delivered. There are at most four kinds of transportation modes to be adopted during the process of transportation. The four kinds of transportation modes are road, rail, water and air. The transportation mode of each trip needs to be decided in order to minimize the total costs and carbon emissions. Owing to the different transportation mode adopt the different route between two adjacent cities, the distance and time between two adjacent cities by the different mode are different. The distances are listed in Table I. The speeds of Road, Rail, Water and Air are set to 60, 100, 30 and 500 *km/h*. The delivery time can be computed by the distance and speed. The cost coefficient by the same mode per unit distance per unit weight is the same which is given in Table II. The transfer cost, time and carbon

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TABLE I. THE DISTANCE BETWEEN TWO ADJACENT CITIES BY DIFFERENT MODES

Route / Mode	A-C	C-D	D-E	E-F	F-G	G-H	H-I	I-J	J-B
Road	160	147	173	256	232	257	394	489	230
Rail	146	131	165	247	206	240	452	480	220
Water	255	400	165	300	220	270	420	400	225
Air	130	122	160	220	200	220	310	385	189

a. The Unit is Kilometer.

TABLE II. THE COST COEFFICIENT OF DIFFERENT MODES

Mode	road	rail	water	air
Cost	0.5	0.2	0.7	1.0

b. The Unit is \$/km•ton.

emissions (Table III) are incurred, and they are different on each city when do the same change. The values of other parameters are set as follow: C_t=7.5(\$/kg), α=1.0(\$/h), β=0.8(\$/h), P=0.4(kg/ton•h), M=30(ton), T_1=20(h), T_2=30(h), O=5500(kg).

B. The Experiment Results

Based on the data given above, the model is solved by ILOG OPL 12.2. The optimal result is given in Table V and the runtime of the experiment is 0.7 second.

The objective value is 53103.0 dollars. For this example, we can notice that the rail and water mode is used frequently.

When the carbon emissions are not considered in the objective and constraint, the optimal solution is shown in Table VI.

The objective value is 24811.6 dollars. We can see from the results in Table VI that the optimal solution is totally different from considering the carbon emissions. Water and rail are the low-carbon transportation mode. When the carbon emissions are considered, the two modes are used more. When the carbon emissions are not considered, the low-cost transportation modes are used more. Water mode is as a better way to supply the rail, but it is not available for the most geographical condition and the cost is expensive in many places. From the view of cost, the water mode is not used.

V. CONCLUSIONS

The intermodal transportation is becoming more popular transportation mode. A mass of transportation make us have to face the negative effect imposed on environment. The cost is not the only criterion to select the transportation mode. So the integer programming model is proposed in this paper to select the transportation mode in each trip with considering the quantity of carbon emissions. The model is solved by the ILOG OPL 12.2 and the optimal solution is obtained for the given example. Carbon emissions are needed to be considered in other intermodal transportation researches to find more reasonable optimal solution not only from the view of cost, but also from the environment. In the near future, the research will be extended to consider carbon emissions in other intermodal transportation researches. The aim is to observe the effect on optimal solution by considering the factor of carbon emissions.

TABLE III. THE COST /TIME/CARBON EMISSIONS ON CITIES WHEN CHANGE MODE

City / Mode change	C	D	E	F	G	H	I	J
Road→Rail	1/0.5/0.2	1/0.3/0.3	2/0.5/0.1	2/0.5/0.2	1/0.2/0.1	2/0.3/0.2	3/0.2/0.3	1/0.2/0.2
Road→Water	2/0.3/0.3	1/0.3/0.3	3/0.3/0.2	1/0.3/0.3	2/0.2/0.3	2/0.2/0.2	2/0.2/0.3	1/0.1/0.3
Road→Air	2/0.4/0.3	2/0.4/0.4	2/0.4/0.3	1/0.3/0.2	1/0.1/0.2	2/0.1/0.3	3/0.3/0.3	1/0.1/0.2
Rail→Road	1/0.3/0.2	1/0.1/0.3	1/0.3/0.3	2/0.1/0.4	3/0.3/0.2	1/0.1/0.2	1/0.1/0.3	2/0.2/0.1
Rail→Water	4/0.2/0.3	3/0.2/0.5	2/0.3/0.3	4/0.5/0.3	5/0.6/0.3	3/0.3/0.2	3/0.3/0.3	2/0.3/0.3
Rail→Air	3/0.3/0.2	3/0.2/0.2	2/0.3/0.2	1/0.1/0.3	3/0.3/0.3	2/0.3/0.3	3/0.3/0.3	2/0.2/0.3
Water→Road	2/0.4/0.1	3/0.2/0.3	1/0.1/0.2	3/0.3/0.1	1/0.2/0.4	1/0.2/0.3	1/0.2/0.3	2/0.4/0.3
Water→Rail	3/0.3/0.3	2/0.3/0.1	2/0.3/0.1	2/0.1/0.3	1/0.1/0.3	3/0.2/0.4	3/0.3/0.2	2/0.2/0.4
Water→Air	5/0.5/0.4	3/0.3/0.1	5/0.6/0.3	7/0.6/0.3	3/0.4/0.2	2/0.3/0.3	4/0.5/0.3	3/0.3/0.3
Air→Road	1/0.2/0.2	1/0.3/0.3	1/0.2/0.2	3/0.3/0.3	1/0.2/0.3	3/0.3/0.1	4/0.4/0.2	2/0.2/0.3
Air→Rail	2/0.3/0.3	3/0.3/0.2	1/0.3/0.3	2/0.3/0.2	3/0.3/0.3	1/0.3/0.3	2/0.2/0.3	2/0.2/0.3
Air→Water	3/0.3/0.3	3/0.3/0.3	4/0.5/0.3	5/0.5/0.3	2/0.2/0.1	4/0.5/0.3	2/0.3/0.3	1/0.1/0.1

c. The Units are \$/*ton*,*h*/*ton*,*kg*/*ton* of Cost, Time and Carbon Emissions On Cities When Change The Mode, Respectively.

TABLE IV. THE COEFFICIENT OF CARBON EMISSIONS OF DIFFERENT MODES

Mode	road	rail	water	air
Carbon emissions coefficient	0.18	0.05	0.03	1.3

d. The unit is kg/ km•ton.

TABLE V. THE OPTIMAL SOLUTION OF THE EXPERIMENT

Route	A-C	C-D	D-E	E-F	F-G	G-H	H-I	I-J	J-B
Mode	Water	Rail	Water	Rail	Water	Rail	Water	Rail	Water

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TABLE VI. THE OPTIMAL SOLUTION OF THE EXPERIMENT WITHOUT CONSIDERING CARBON EMISSIONS

Route	A-C	C-D	D-E	E-F	F-G	G-H	H-I	I-J	J-B
Mode	Rail	Road	Rail	Road	Rail	Road	Rail	Road	Rail

REFERENCES

[1] H. Min, "International intermodal choices via chance-constrained goal programming," Transportation Research Part A: General, vol. 25, pp. 351-362, 1991.

[2] V. R. Reddy and R. G. Kasilingam, "Intermodal transportation considering transfer costs," in Proceedings of the 1995 Global Trends Conference of the Academy of business Administration: Aruba, 1995.

[3] R. Van Duin and H. Van Ham, "A three-stage modeling approach for the design and organization of intermodal transportation services," in IEEE International Conference on Systems, Man, Cybernetics. vol. 4, pp. 4051-4056, 1998.

[4] P. Arnold, D. Peeters, and I. Thomas, "Modelling a rail/road intermodal transportation system," Transportation Research Part E: Logistics and Transportation Review, vol. 40, pp. 255-270, 2004.

[5] T. Chang, "Best routes selection in international intermodal networks," Computers & operations research, vol. 35, pp. 2877-2891, 2008.

[6] Y. Bontekoning, C. Macharis, and J. Trip, "Is a new applied transportation research field emerging?--A review of intermodal rail-truck freight transport literature," Transportation Research Part A: Policy and Practice, vol. 38, pp. 1-34, 2004.

[7] C. Macharis and Y. Bontekoning, "Opportunities for OR in intermodal freight transport research: A review," European Journal of Operational Research, vol. 153, pp. 400-416, 2004.

[8] K. Ahn and H. Rakha, "The effects of route choice decisions on vehicle energy consumption and emissions," Transportation Research Part D: Transport and Environment, vol. 13, pp. 151-167, 2008.

[9] H. Ren, W. Zhou, K. Nakagami, W. Gao, and Q. Wu, "Multi-objective optimization for the operation of distributed energy systems considering economic and environmental aspects," Applied Energy, vol. 87, pp. 3642-3651, 2010.

[10] S. Cholette and K. Venkat, "The energy and carbon intensity of wine distribution: A study of logistical options for delivering wine to consumers," Journal of Cleaner Production, vol. 17, pp. 1401-1413, 2009.

[11] B. Sundarakani, R. de Souza, M. Goh, S. Wagner, and S. Manikandan, "Modeling carbon footprints across the supply chain," International Journal of Production Economics, vol. 128, pp. 43-50, 2010.

[12] C. Liao, P. Tseng, and C. Lu, "Comparing carbon dioxide emissions of trucking and intermodal container transport in Taiwan," Transportation Research Part D: Transport and Environment, vol. 14, pp. 493-496, 2009.

[13] N. Kim, M. Janic, and B. van Wee, "Trade-off between carbon dioxide emissions and logistics costs based on multiobjective optimization," Transportation Research Record: Journal of the Transportation Research Board, vol. 2139, pp. 107-116, 2009.

[14] "ILOG optimization suit white paper," 2001.

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